

Learned versus Exact Gauge Invariance in Graph Neural Networks for Lattice Gauge Theory

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Neural-network surrogates for lattice gauge theory must respect gauge symmetry, and the field has split between architectures that build the symmetry in exactly and flexible architectures expected to learn it from data. We test the learnable side of this divide directly in two-dimensional compact U(1) lattice gauge theory. Three graph encodings of the same gauge configurations—link variables as field nodes, link variables as edge features, and a gauge-invariant plaquette oracle—are trained under an identical protocol to regress the Wilson action, Wilson loops, and the topological charge. The two link-level encodings fail completely: predictions are uncorrelated with every gauge-invariant target ($|r| \lesssim 0.1$), robustly across couplings, volumes, parameter budgets, and training variations, while the oracle reaches $r \approx 0.999$. A direct measurement of the gauge-invariance error ϵ_{gauge} —the ratio of prediction variation along a gauge orbit to variation across configurations—shows why: the link-level models sit at $\epsilon_{\text{gauge}} \approx 1$ everywhere, i.e. their outputs contain no gauge-invariant component at all, whereas the oracle achieves $\epsilon_{\text{gauge}} = 0$ exactly. Retraining with random gauge transformations as data augmentation does not move either number at any training-set size; it merely shrinks the network output toward a constant, while the oracle already reaches $r \approx 0.87$ from only 50 configurations. At this model scale, gauge invariance is not learned at any price we could pay; it must be supplied, through invariant inputs or equivariant architecture.

I. INTRODUCTION

Machine-learned surrogates are increasingly used inside lattice field theory workflows: as generative proposal distributions for exactness-preserving samplers [1–3], as part of a broader program coupling lattice field theory to probabilistic machine learning [4], and as general-purpose emulators of local theories [5–8]. For gauge theories, every such application confronts the same design decision. Gauge symmetry is a *redundancy*: physical observables are constant on gauge orbits that occupy almost the entire configuration space. An architecture can either enforce this exactly—through gauge-invariant input features or gauge-equivariant network operations [2, 3, 5–7]—or present raw link variables to a flexible model and rely on the data to teach the symmetry.

The trade-off is usually argued qualitatively. Exact constructions guarantee the symmetry but restrict the architecture to the group fixed at design time; flexible architectures port across theories, boundary conditions, and discretizations, but their symmetry error is rarely measured. In a companion study of scalar ϕ^4 theory [8] we found the flexible side of this argument convincing for *spacetime* symmetries: a heterogeneous bipartite graph neural network (GNN) with no built-in translation symmetry learned it to high accuracy, transferred across lattice volumes, and matched parameter-matched convolutional baselines. That paper deferred a sharper question to this one: does the same flexibility-over-rigidity argument survive contact with *gauge* symmetry?

This paper answers that question quantitatively, and the answer is no. Working in quenched two-dimensional

compact U(1) lattice gauge theory—small enough that every experiment is exactly validated and cheaply repeatable—we train one GNN architecture on three graph encodings of the same Monte Carlo ensembles (Fig. 1): link variables as first-class field nodes (Variant A), link variables as directed edge features (Variant B), and precomputed plaquette features, i.e. gauge-invariant inputs, as a control (Variant C). The targets are the Wilson action, Wilson loops of five sizes, and the topological charge—all exactly gauge invariant.

Our contributions are threefold. First, a *null result with unusual robustness*: Variants A and B are statistically indistinguishable from chance on every gauge-invariant target, across couplings, volumes, seeds, matched parameter budgets, longer training, lower learning rates, single-task objectives, and deeper message passing, while Variant C is at ceiling under the identical protocol (Sec. IV A). Second, a *direct measurement of learned invariance*: we define a per-target gauge-invariance error ϵ_{gauge} from the spread of predictions over random gauge orbits and show that A and B sit at $\epsilon_{\text{gauge}} \approx 1$ everywhere—their outputs are pure functions of the gauge representative—while C achieves $\epsilon_{\text{gauge}} = 0$ exactly, by a bit-level argument about its inputs (Sec. IV B). Third, an *augmentation study*: retraining A and B with fresh random gauge transformations of every training example, across training-set sizes spanning two orders of magnitude, changes neither accuracy nor ϵ_{gauge} ; the only effect is a collapse of the network output toward a constant (Sec. IV C). The oracle, by contrast, reaches $r = 0.87 \pm 0.06$ on the action from 50 configurations.

The interpretation is elementary but, we believe, worth establishing experimentally: a single link variable is *exactly* uninformative about any gauge-invariant quantity,

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because a random gauge transformation maps it to a uniformly distributed angle regardless of the underlying physics (Sec. V). Invariant information first appears in oriented four-link products—plaquettes—so any learning signal is invisible at first order in feature–target correlations, for the supervised loss and for the augmentation-consistency loss alike. The resulting optimization problem resembles learning a parity function: possible in principle, but with no gradient to follow at realistic scale. Gauge invariance, unlike translation invariance, is not an inductive bias a flexible model acquires from data; it is a precondition for the data to carry usable signal at all.

II. SETUP

Theory and ensembles. We work with quenched compact U(1) gauge theory on periodic $L \times L$ lattices, with link angles $\theta_\mu(x) \in (-\pi, \pi]$, $U_\mu(x) = e^{i\theta_\mu(x)}$, and the Wilson action [9]

$$S[\theta] = \beta \sum_x [1 - \cos \theta_P(x)], \quad (1)$$

where the plaquette angle is $\theta_P(x) = \theta_1(x) + \theta_2(x + \hat{e}_1) - \theta_1(x + \hat{e}_2) - \theta_2(x)$. Ensembles of 4000 decorrelated configurations are generated per (β, L) point by a standard heatbath algorithm, for $\beta \in \{0.5, 1, 2, 3, 4\}$ and $L \in \{8, 16, 32\}$, and validated against the exact finite-volume mean plaquette of the 2D theory from the character expansion $[\langle \cos \theta_P \rangle \rightarrow I_1(\beta)/I_0(\beta)]$ as $V \rightarrow \infty$.

Targets. Each configuration is labeled with three families of exactly gauge-invariant quantities: the total action Eq. (1); per-configuration Wilson loops $W(R, T)$, the lattice average of cos of the oriented boundary angle sum of $R \times T$ rectangles, for $(R, T) \in \{(1, 1), (2, 2), (2, 4), (3, 3), (4, 4)\}$; and the topological charge $Q = \frac{1}{2\pi} \sum_x \widehat{\theta}_P(x)$, with $\widehat{\theta}_P$ wrapped to $(-\pi, \pi]$, an exact integer on the torus. Wilson-loop targets are the raw per-configuration values (not $-\ln W$, which is undefined on the large fraction of configurations with $W \leq 0$ at large loop area). All targets are z -scored with training-split statistics for loss balance; scale-dependent metrics are computed after inverting the standardization, and Pearson r is unaffected by it.

Model. The architecture is the heterogeneous bipartite GNN of Ref. [8], built on typed message passing [10–12] in PYG [13]: spacetime nodes carry a constant feature (all geometry lives in displacement edge features, which restores exact translation invariance), field content enters through dedicated node or edge encoders, and B residual message-passing blocks with layer normalization propagate information before task heads read the result. The action head sums a per-site MLP over spacetime nodes; Wilson-loop and charge heads mean-pool all node types and apply an MLP, one head per target. The coupling β is appended to the input features

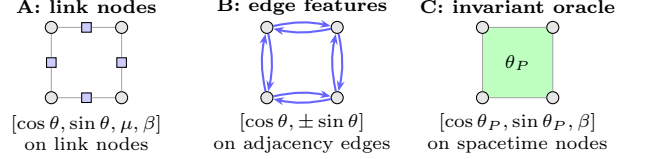


FIG. 1. Three graph encodings of the same U(1) configuration (one plaquette shown). **A**: each of the $2L^2$ links is a field node attached to its two endpoint sites by typed edges; message passing alternates link→site, site→site, site→link. **B**: links ride as features on the directed site-site adjacency edges, with the reversed edge carrying the inverse link $[\cos \theta, -\sin \theta]$. **C** (control): each site carries the trigonometric features of its plaquette—inputs that are gauge invariant *by construction*. A plaquette is a 4-link neighborhood: in A and B its information becomes available within two message-passing blocks.

and to every head. Training follows the frozen protocol of Ref. [8]: AdamW, cosine schedule, 150 epochs, batch 32, hidden width $H = 64$, $B = 3$ blocks, an 80/10/10 train/validation/test split, and 3–5 random seeds per cell.

III. THREE GRAPH ENCODINGS OF GAUGE LINKS

The experiment hinges on presenting identical physics to identical learning machinery under encodings that differ only in *where the gauge redundancy lives* (Fig. 1).

Variant A (link nodes) is the natural extension of the fields-as-nodes thesis of Ref. [8]: every link is a field node with features $[\cos \theta, \sin \theta, \text{onehot}(\mu), \beta]$, connected to its endpoint sites by typed directed edges, with a three-stage message-passing block (link→site, site→site, site→link).

Variant B (edge features) attaches $[\cos \theta, \sin \theta]$ to the existing directed site-site adjacency edges, alongside the displacement features; the reversed edge carries the inverse link $[\cos \theta, -\sin \theta]$. Only spacetime nodes remain.

Variant C (invariant oracle) precomputes the one independent plaquette per site in 2D and places $[\cos \theta_P, \sin \theta_P, \beta]$ on the spacetime nodes. This is the information-level ceiling: the model is handed gauge invariance and asked only to compose invariants into targets. It is a control, not a proposal—in higher dimensions or for larger structures, enumerating all invariants is exactly the expense one hopes a learned model avoids.

At fixed width $H = 64$, Variant A carries $2.5\times$ the parameters of B and C (296k vs. 118k), because of its extra encoder and per-block message functions. We therefore run the comparison in both a fixed- H arm and a parameter-matched arm (B, C widened to $H = 102$ to match A’s budget; A narrowed to $H = 40$ to match B’s).

IV. RESULTS

A. Learned invariance does not happen

Table I shows the headline comparison at the three (β, L) cells where all variants were trained. Variants A and B are statistically indistinguishable from chance on the action, all five Wilson-loop sizes, and the topological charge; Q “accuracy” equals the base rate of the modal charge sector. Variant C, trained under the identical protocol, reaches $r \geq 0.997$ on the action and short-distance loops.

We stress-tested the null before accepting it. It persists at every A/B cell trained ($\beta \in \{1, 2\}$, $L \in \{8, 16\}$, three seeds each), while C’s ceiling holds over its full matrix ($\beta \in \{0.5, \dots, 4\}$ at $L \in \{8, 16\}$; $\beta \in \{1, 2, 4\}$ at $L = 32$); under both directions of parameter matching (max $|r| = 0.062$ on the action); and under four targeted escape hatches for Variant A— $3\times$ longer training, a lower learning rate with $2\times$ training, a single-task objective on the most local target $W(1\times 1)$ alone, and a deeper $B = 6$ stack (max $|r| = 0.041$ on the action, 0.094 over all targets). Training loss decreases in every case, but only to the constant-predictor floor: the networks converge to predicting the per-target mean, not to fitting individual training configurations, and no generalizing signal appears.

The single systematic exception is at $\beta = 1$, where both variants show a small but seed-consistent action correlation (A: $r = 0.059 \pm 0.013$, B: $r = 0.083 \pm 2 \times 10^{-4}$). We return to it below; it is the exception that maps the boundary of the null.

B. The gauge-invariance error

Accuracy alone cannot distinguish “failed to learn the physics” from “failed to learn the symmetry.” We therefore measure invariance directly. For each test configuration θ_i we draw $K = 32$ random gauge transformations $\theta_\mu(x) \rightarrow \theta_\mu(x) + \alpha(x) - \alpha(x + \hat{e}_\mu)$ with i.i.d. $\alpha(x) \sim \mathcal{U}(-\pi, \pi]$, and define, per prediction head $\hat{y}$,

$$\epsilon_{\text{gauge}} = \frac{\langle \text{std}_g \hat{y}(g \cdot \theta_i) \rangle_i}{\text{std}_i \hat{y}(\theta_i)}, \quad (2)$$

the mean prediction spread along gauge orbits in units of the prediction spread across configurations. A gauge-invariant predictor has $\epsilon_{\text{gauge}} = 0$; a predictor that responds to the gauge representative as strongly as to the physics has $\epsilon_{\text{gauge}} \approx 1$. The ratio is invariant under affine rescaling of $\hat{y}$, so it can be evaluated on the model’s standardized output scale.

Two protocol details matter. First, gauge copies are generated in float64 from the float32-stored links; the resulting $\sim 10^{-15}$ transformation noise in $\cos \theta_P$, $\sin \theta_P$ is seven orders of magnitude below float32 feature resolution, so Variant C’s *inputs are bit-identical* across

each orbit. Its $\epsilon_{\text{gauge}} = 0$ rows in Table II are exact, a built-in sanity anchor for the pipeline. Second, achieving that exactness requires evaluating each graph unbatched: batched message passing is position-in-batch dependent at float32 machine precision ($\sim 10^{-8}$ output differences between bit-identical graphs at different batch positions), which would otherwise put a spurious floor under the oracle’s zero.

Table II gives the result. Variants A and B sit at $\epsilon_{\text{gauge}} \approx 1$ on every head: their predictions vary as much when the gauge is changed at fixed physics as when the physics itself changes. The trained functions contain *no gauge-invariant component*. This is a strictly stronger statement than the accuracy null—these models did not learn a noisy approximation to an invariant observable; they learned nothing invariant at all. The same holds for the parameter-matched arm ($\epsilon_{\text{gauge}} \in [0.95, 1.04]$) and all four escape hatches ($\epsilon_{\text{gauge}} \in [0.98, 1.06]$), over 148 checkpoints in total; every one of the 36 Variant-C checkpoints returns exactly zero.

The $\beta = 1$ exception is visible here too, from the other side: both link-level variants dip a few percent below unity (B: $\epsilon_{\text{gauge}} = 0.968 \pm 0.002$ on the action head; A: 0.963 ± 0.026 on the $W_{1\times 1}$ head)—the size expected from a percent-level invariant component, consistent with the small r observed at that coupling. We interpret this as a weak-coupling foothold: at small β the plaquette distribution is broad, and low-order combinations of nearby links carry slightly more usable signal relative to the gauge noise. The foothold does not grow with data or training in any experiment we ran.

C. Augmentation does not buy invariance

If the missing ingredient were merely an invariance *prior*, the standard remedy would be data augmentation: draw a fresh random gauge transformation of every training configuration at every access, so the model sees each physical state under an ever-changing gauge and the loss itself penalizes gauge dependence. The labels, being exactly invariant, transfer unchanged; augmentation costs nothing but compute.

Figure 2 shows the outcome across training-set sizes $n \in \{50, \dots, 3200\}$, with matched unaugmented base-lines and the oracle as reference. Augmentation changes nothing that matters. Test correlations remain at chance for both variants at every n (and in a full-size check at $L = 16$); ϵ_{gauge} remains at unity for every head at every n . What augmentation *does* do—visible only in the components of Eq. (2)—is drive the network toward the trivial invariant: the constant function. The config-to-config spread of the augmented Variant-A action head falls from 1.4×10^{-3} (standardized units, $n = 50$) to 1.1×10^{-6} ($n = 3200$); the unaugmented arm follows the same trajectory (2.5×10^{-3} to 1.6×10^{-6}) as ordinary training regresses an unlearnable target toward its mean. In both cases the *residual* fluctuation is pure gauge noise—which

TABLE I. A/B/C comparison on gauge-invariant targets (test split, mean $\pm$ std over seeds; 3 seeds for A/B, 5 for C). Q acc. is exact-integer accuracy after rounding. Variants A and B are at chance on every target, up to the percent-level $\beta = 1$ correlation discussed in the text; the invariant oracle C is at ceiling except for the receptive-field-limited 4×4 loop (Sec. IV D).

Variant	L	β	Params	Action r	$W_{1 \times 1} r$	$W_{4 \times 4} r$	Q acc.
A	8	1	296k	0.059 ± 0.013	0.080 ± 0.005	0.065 ± 0.009	0.255 ± 0.000
A	8	2	296k	-0.017 ± 0.027	-0.004 ± 0.016	-0.015 ± 0.033	0.373 ± 0.000
A	16	2	296k	0.036 ± 0.105	-0.012 ± 0.028	0.050 ± 0.026	0.145 ± 0.000
B	8	1	118k	$0.083 \pm 2 \times 10^{-4}$	0.086 ± 0.001	0.040 ± 0.008	0.255 ± 0.000
B	8	2	118k	-0.057 ± 0.002	-0.060 ± 0.006	-0.008 ± 0.057	0.373 ± 0.000
B	16	2	118k	0.014 ± 0.012	0.046 ± 0.012	0.030 ± 0.036	0.145 ± 0.000
C	8	1	118k	$1.000 \pm 6 \times 10^{-5}$	$0.999 \pm 2 \times 10^{-4}$	0.021 ± 0.026	0.984 ± 0.003
C	8	2	118k	$0.999 \pm 3 \times 10^{-5}$	$0.999 \pm 9 \times 10^{-5}$	0.476 ± 0.011	0.997 ± 0.001
C	16	2	118k	$0.999 \pm 7 \times 10^{-5}$	0.998 ± 0.001	0.316 ± 0.009	0.981 ± 0.007

TABLE II. Gauge-invariance error ϵ_{gauge} [Eq. (2), $K = 32$ gauge copies per test configuration] by target: total action S , Wilson loops, topological charge Q . Mean $\pm$ std over seeds. Variant C’s zeros are exact (bit-identical inputs; see text). Across all 50 measured cells—including parameter-matched budgets, escape hatches, and every augmentation arm—A and B lie in $\epsilon_{\text{gauge}} \in [0.95, 1.06]$.

Variant	L	β	$\epsilon_{\text{gauge}}^S$	$\epsilon_{\text{gauge}}^{W_{1 \times 1}}$	$\epsilon_{\text{gauge}}^{W_{4 \times 4}}$	$\epsilon_{\text{gauge}}^Q$
A	8	1	0.990 ± 0.033	0.963 ± 0.026	0.959 ± 0.020	0.973 ± 0.019
A	8	2	1.000 ± 0.013	0.989 ± 0.009	1.015 ± 0.012	0.996 ± 0.003
A	16	2	1.002 ± 0.018	1.008 ± 0.009	1.028 ± 0.007	1.011 ± 0.005
B	8	1	0.968 ± 0.002	0.976 ± 0.005	0.967 ± 0.009	1.005 ± 0.045
B	8	2	1.012 ± 0.005	1.026 ± 0.010	1.020 ± 0.022	1.008 ± 0.029
B	16	2	1.005 ± 0.031	1.026 ± 0.017	1.047 ± 0.028	1.033 ± 0.021
C	8	1	0	0	0	0
C	8	2	0	0	0	0
C	16	2	0	0	0	0

is why the ratio ϵ_{gauge} stays pinned at unity while both of its ingredients collapse by three orders of magnitude. Optimization can damp the gauge-covariant component of the function; at this scale it never rotates it into the invariant subspace.

The contrast with the oracle’s data efficiency quantifies what exact invariance is worth: Variant C reaches $r = 0.87 \pm 0.06$ on the action with 50 training configurations—fewer samples than the model has layers of width 64—and is at ceiling by a few hundred. Handing the model the invariant is worth more than any amount of augmented data we could supply.

D. What the oracle does with the same budget

Variant C’s failures are as informative as its successes, because they calibrate what even an invariant-input model cannot do at this scale. Two systematic effects appear (Figs. 3 and 4).

Receptive field. Prediction quality for $W(R, T)$ falls with loop area at fixed depth— $r = 0.316 \pm 0.009$ for the 4×4 loop at $B = 3$, $L = 16$, $\beta = 2$ with the full training set—and, in the disordered phase, recovers with added depth, tracking the number of lattice spacings a node can

see. This turns the locality interpretation of Ref. [8] into a measured scaling and sets the depth budget a Phase-scale application must plan for.

Ordered-phase over-smoothing. At $\beta = 4$ the input feature spread is roughly halved (the plaquette distribution sharpens), and deeper stacks *hurt*: global readouts collapse to base rate and the large-loop correlation inverts its depth trend. The diagnostic in Fig. 4 attributes this to a trained-in representation collapse—post-encoder node embeddings become nearly identical across sites, so pooled readouts retain only graph-mean information (the bulk action survives; per-site structure and rare-event precision do not). Residual connections and layer normalization, which prevented this failure in the scalar theory [8], do not prevent it here.

Finally, exact-integer topological charge accuracy obeys a rounding floor: given a regression correlation q_r and charge scale σ_Q , accuracy tracks $P(|\mathcal{N}(0, \sigma_Q \sqrt{1 - q_r^2})| < \frac{1}{2})$, which hardens with volume as σ_Q grows. Reported Q accuracies should be read jointly with q_r against this floor.

A-5 gauge augmentation: accuracy and eps_gauge vs training size (u1_L8_beta2, K=32 copies)

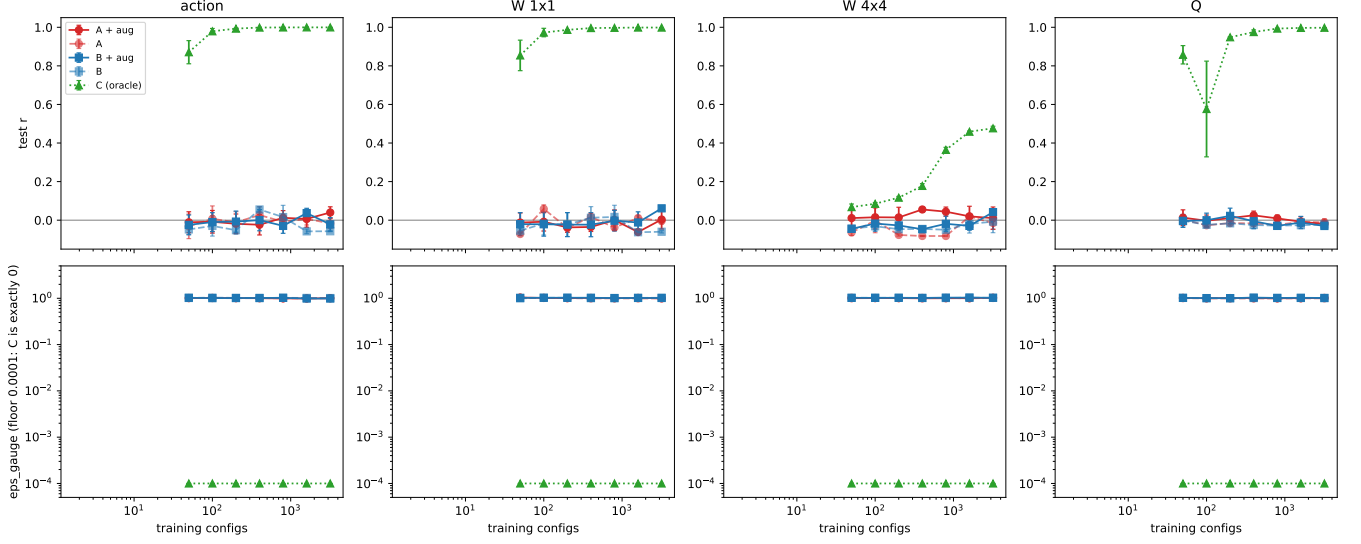


FIG. 2. Gauge-transform data augmentation does not rescue the link-level encodings ($L = 8$, $\beta = 2$; 3 seeds per point; the C $n=3200$ reference uses the 5-seed protocol runs). **Top**: test Pearson r vs. training-set size for the action, two Wilson-loop sizes, and the topological charge. Variants A and B, with (solid) and without (dashed) on-the-fly random gauge augmentation, remain at chance at every size ($|r| \leq 0.088$ across all arms, sizes, and targets), while the invariant oracle C (green) reaches $r = 0.87 \pm 0.06$ on the action from 50 configurations and $0.999 \pm 3 \times 10^{-4}$ by 400. **Bottom**: ϵ_{gauge} for the same runs stays at unity for A and B at every size—augmentation never produces an invariant component; C is exactly zero (plotted at the 10^{-4} axis floor).

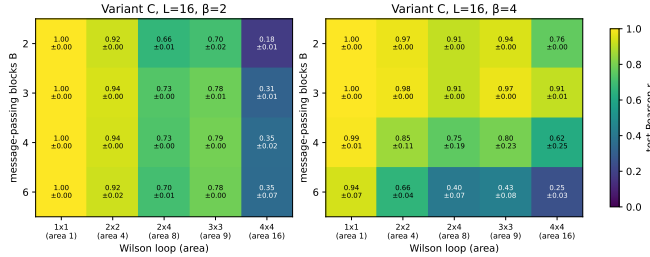


FIG. 3. Receptive-field study on the invariant oracle (Variant C, $L = 16$): test r for Wilson loops of increasing area vs. message-passing depth B , at $\beta = 2$ (disordered) and $\beta = 4$ (ordered). Depth extends the reachable loop size at $\beta = 2$ but degrades large-loop readouts at $\beta = 4$ (the Q readout of the same runs collapses to base rate at $B = 6$; see text).

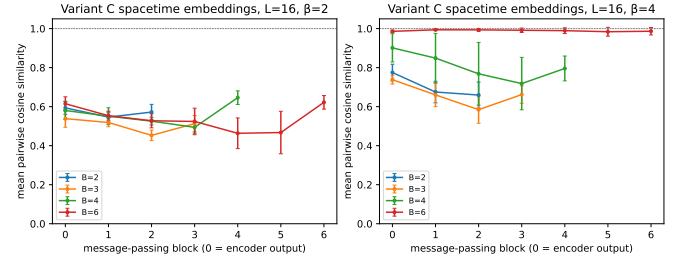


FIG. 4. Over-smoothing diagnostic on trained oracle models ($\beta = 2$ and $\beta = 4$ panels): mean pairwise cosine similarity of spacetime-node embeddings after each message-passing block. Deep stacks at $\beta = 4$ collapse the node representations (two seeds reach identical embeddings), while $\beta = 2$ models stay healthy at every depth; per-seed collapse predicts per-seed readout failure.

V. WHY THE LEARNING SIGNAL VANISHES

The completeness of the failure has an elementary explanation. Under a gauge transformation with i.i.d. uniform $\alpha(x)$, each individual link angle is shifted by $\alpha(x) - \alpha(x + \hat{e}_\mu)$, which is itself (marginally) uniform on the circle. The gauge-orbit marginal of any *single* link variable is therefore the uniform distribution, independent of the physical configuration: one link carries exactly zero information about any gauge-invariant quantity. Invariant information first appears in the oriented product of four links around a plaquette—precisely the

feature Variant C precomputes.

For learning dynamics this has a specific consequence. At initialization, the gradient of the loss with respect to early-layer weights is built from correlations between low-order functions of the inputs and the targets; for A and B those correlations vanish identically, not approximately, by the argument above. In pilot probes, the empirical action correlation of single-link features is statistically zero, and a plain MLP given *all* raw link features of a configuration memorizes its training set (train $r \approx 1$) with zero test generalization, while the same MLP on plaquette features generalizes immediately [TODO: com-

mit probe script for exact numbers]. The task is thus structurally analogous to learning a parity: the target is a specific high-order combination of inputs, all low-order marginals are uninformative, and gradient-based training must find the combination without first-order guidance. Message passing does not remove this barrier; it only makes the four-link combination *representable* within two blocks, not *discoverable*.

The same argument explains the augmentation null. The augmentation-consistency signal—penalizing $\hat{y}(g \cdot \theta) \neq \hat{y}(\theta)$ —is mediated by the same input-output correlations, so it too vanishes at first order. The one direction in which it *does* act is the trivial one: shrinking the overall output scale toward a constant, which is exactly what we observe (Sec. IV C). And the $\beta = 1$ foothold fits the picture from the other side: where the plaquette distribution is broadest, the lowest-order non-vanishing (four-link) correlations are largest relative to the gauge noise, and a percent-level invariant component appears—but with no path to grow, since each additional order of combination reinstates the same barrier.

VI. RELATED WORK

Exactly gauge-equivariant and gauge-invariant architectures are now standard in lattice ML: equivariant normalizing flows for U(1) and SU(N) sampling [2, 3], lattice gauge equivariant CNNs [5], gauge-equivariant networks for quantum lattice models [6], and, closest to our setting, gauge-equivariant message passing on graphs [7]. These works *construct* the symmetry and measure the resulting physics; none of them, to our knowledge, measures what happens when the symmetry must be learned—the counterfactual that justifies the construction. Our contribution is that counterfactual, run under a controlled same-architecture, same-data comparison, with a quantitative invariance error. Our Variant C is deliberately *not* an equivariant architecture but an invariant-input preprocessing step: it bounds from above what any symmetry-supplying method can gain at this scale, while remaining architecture-agnostic. General graph learning work [10–12] provides the typed message-passing machinery; Ref. [8] develops the heterogeneous fields-as-nodes formulation whose gauge extension is tested here; and Ref. [4] exemplifies the observable-regression setting our targets are drawn from.

VII. DISCUSSION AND OUTLOOK

The boundary of flexibility-over-rigidity. The companion scalar-theory study [8] argued that spacetime symmetries need not be hard-wired: a flexible heterogeneous GNN learned translation invariance from data and transferred across volumes. This paper locates the boundary of that argument. The distinction is not “symmetries are learnable” versus “symmetries are not”; it is

whether the symmetry *leaves local features informative*. Translations act on the lattice while leaving each field value a meaningful physical quantity; gauge transformations act on the fields themselves and randomize every local feature completely. In the first case a flexible model sees signal it can refine toward symmetry; in the second there is no signal until the model has already found the invariant combinations—a chicken-and-egg problem that neither data volume, nor training time, nor explicit augmentation resolved at any point in our scans.

Implications for surrogate design. For the next stage of this program—a surrogate for the fermion determinant of the Schwinger model, where the target is genuinely expensive—these results fix the design: gauge information must enter through invariant inputs (plaquettes and, where needed, longer Wilson lines) or through an exactly equivariant message-passing layer [7]; raw-link flexibility is not an option to be traded off, but a dead end. The receptive-field and over-smoothing measurements (Sec. IV D) additionally set the depth budget and flag the ordered phase as the regime where deep stacks need architectural care.

Limitations. All results are at laptop scale by design: 2D, quenched, $L \leq 32$, models of $\lesssim 300k$ parameters, a single frozen training protocol, and $\lesssim 3200$ training configurations per cell. We cannot exclude that very large models, very long training, or curriculum schedules (e.g. exploiting the $\beta = 1$ foothold) eventually crack the parity-like barrier—for parities themselves, scale sometimes does. What the experiments establish is the shape and the robustness of the barrier at the scale where lattice surrogates currently operate, and the extreme cheapness of the alternative: Variant C costs one plaquette computation per site and 50 training configurations to beat everything Variants A and B achieved with two orders of magnitude more data.

VIII. CONCLUSION

We asked whether a flexible graph neural network can learn gauge invariance the way it demonstrably learns spacetime symmetries, and measured the answer in 2D compact U(1) lattice gauge theory. It cannot, at any scale we probed: link-level encodings produce predictions with no gauge-invariant component ($\epsilon_{\text{gauge}} \approx 1$ on every target, in every experiment), gauge-transform augmentation moves them only toward the trivial constant function, and the same architecture handed gauge-invariant inputs is at ceiling from 50 training examples. The lesson we draw is sharp and, we believe, portable: symmetries that annihilate local information—gauge symmetries—must be supplied to a learned surrogate, not asked of it. How best to supply them at scale, by invariant preprocessing or equivariant construction, is the right question for the next phase of this program.

ACKNOWLEDGMENTS

All experiments are reproducible from committed scripts and configuration files in the project repository; every number in this paper regenerates from logged run records. [TODO: repository URL/DOI before submission]

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